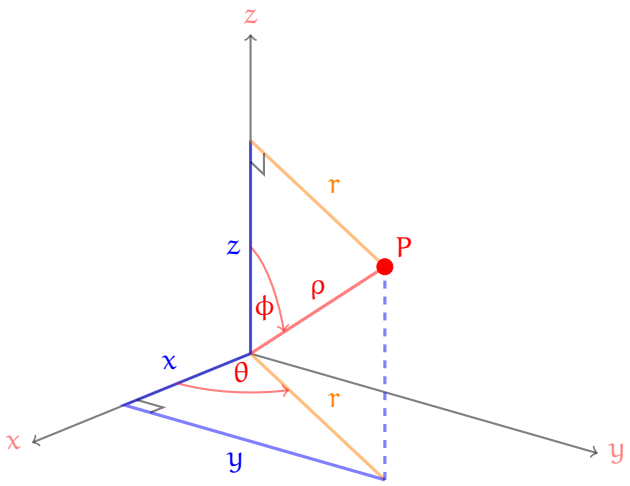
A. **Spherical Conversion.** We convert from spherical to other coordinates.



[Spherical ↔ Cylindrical] and [Spherical ↔ Standard]

$r =$

$x =$

$y =$

$z =$

$\rho^2 =$

and thus when ρ is nonnegative: $\rho =$

$z \sin \phi =$

and thus when z is nonzero: $r/z =$

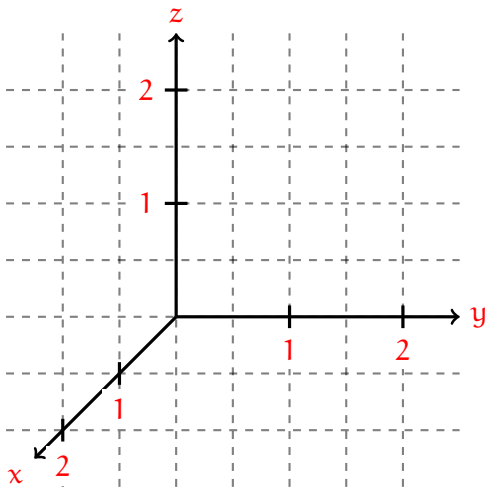
and thus when r is nonzero: $z/r =$

Example 1. Convert the cylindrical equation to spherical, then sketch.

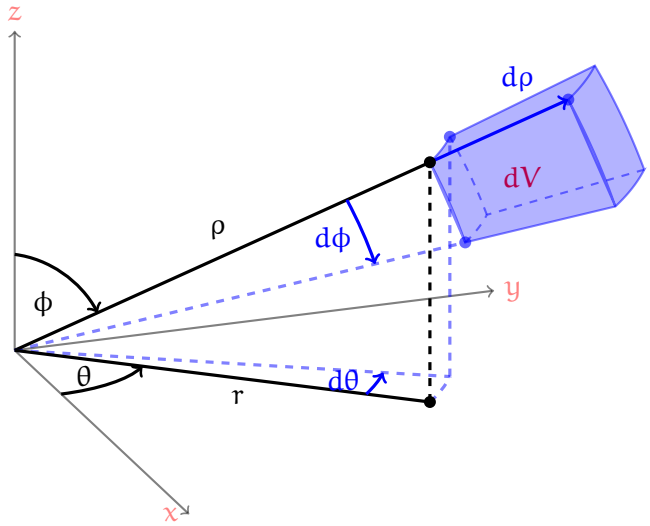
$z = \sqrt{3}r$ with $r \geqslant 0$

Solution. First we rewrite the equation using conversion formulas.

$z = \sqrt{3}r$



B. **Spherical Integration.** Imagine the infinitesimal bit of volume dV obtained from point with coordinates ρ, ϕ, θ by increasing ρ by $d\rho$, ϕ by $d\phi$, and θ by $d\theta$.



We can approximate dV by a rectangular box with dimensions

[length] =

[width] =

[height] =

In spherical coordinates $dV =$

Example 2. Use spherical coordinates to derive the volume of a radius R ball

$$x^2 + y^2 + z^2 \leqslant R^2$$

We can also view in Desmos.

Solution. A solid ball of radius R centered at the origin is given by inequalities

Thus: [volume of ball] $=$

Example 3. Let E be the region outside the cone $z = \sqrt{x^2 + y^2}$ and inside the upper half of the ball $x^2 + y^2 + z^2 \leqslant 4$. Use order $d\rho d\phi d\theta$ to find

$$\iiint_E \frac{1}{\sqrt{x^2 + y^2 + z^2}}\, dV$$

Solution. First we convert the bounding surfaces to spherical coordinates.

$x^2 + y^2 + z^2 = 4$

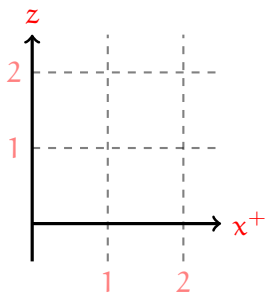
$z = \sqrt{x^2 + y^2}$

Neither equation involves:

so the region is obtained by rotating its x^+z–cross section

x^+ means the positive x–direction.

Let's sketch its x^+z cross–section.



Given fixed θ from $0 \leqslant \theta \leqslant 2\pi$, the range of ϕ is:

Given fixed (θ, ϕ) from the $\theta\phi$–shadow, the range of ρ is:

Thus: $\iiint_E \frac{1}{\sqrt{x^2 + y^2 + z^2}}\, dV =$